

GTOOnline System Overview

GTOOnline is a simple social networking application with a set of basic features similar to those found on sites like Facebook and LinkedIn. GTOOnline users maintain a basic profile including their name, birthdate, hometown, current city, schools they have attended, and jobs held. Users connect to other users by sending friend requests. If a request is accepted, a friendship link is established between the two users. In addition to maintaining basic profile information, users can let their friends know what they are currently doing by writing status updates. Users can also comment on status updates.

GTOOnline also supports admin users who can run reports that aggregate certain information across the entire social network. The system should maintain the **Last Login** date and time of each admin.

Logging In to GTOOnline

Figure 1 shows the GTOOnline login screen. All users are uniquely identified by their **Email Address**. Providing a valid **Email Address** and **Password** combination will log the user into the system. Providing invalid login credentials should display an error message and return to the login screen.



Figure 1 - GTOOnline Login Screen

Users who are new to GTOOnline must register first. A *Register* button is provided directly on the login page. Clicking this button displays the new user registration form. Figure 2 shows the new user registration form. All fields on this form are required and cannot be changed at a later time. (Users who forget their passwords are just plain out of luck!)

After the user clicks **Register**, the system should verify that all fields are filled, that the **Email**

Address has not already been registered, and that the **Password** and **Confirm Password** fields are equal. If any validations fail, the user should be provided with meaningful error messages to make corrections.

GTOOnline New User Registration

GTOOnline  *Lite Social Networking
for Networking Socialites*

First Name

Last Name

Email

Password

Confirm Password

[Returns to the login screen](#)

All fields are required

Figure 2 - New User Registration Form

GTOOnline User Profiles

All GTOOnline users (except administrative users) have a profile containing basic information. After a new user registers with GTOOnline, they should be taken immediately to the Edit Profile screen. This interface is shown in Figure 3. The basic profile properties include the user's **Sex**, **Birthdate**, **Current City**, **Hometown**, and any number of **Interests**.

The profile also contains information about the user's education. The set of available **Schools** and their **Types** is maintained behind the scenes by the database administrator, so the user will select his or her school from a fixed list, instead of typing it in explicitly. A user can have any number of schools associated with his or her profile and can also provide a **Graduation Date** for each school. If a user has not yet graduated, the graduation date can be left blank. The interface should allow the user to add multiple schools and also delete schools from his or her profile in case of an error. It is also possible that the same school will appear multiple times with different graduation dates.

The profile also contains professional information. A list of employers is maintained behind the scenes by the database administrator. The user will select his or her **Employer** from the list and then provide a **Job Title**. The **Job Title** field can be any value provided by the user—i.e., there is no preset list of job titles. A profile can contain multiple **Employers** and the same **Employer** may even appear multiple times as long as the **Job Title** is different in each case. The interface also allows the user to delete associated employers in case of an error.

The only required fields on the profile are **Sex** and **Birthdate**. All the other fields may be left blank (e.g., no **Current City**, **Hometown**, **School**, or **Employer** information).

Edit GTOOnline Profile for Michael Bluth

Michael Bluth

Sex Male

Birthdate 1968-06-20

Current City

Hometown

Interests Tennis
Watching Inception Over and Over Again
Seinfeld

Add

Education **Add Another School**

School Georgia Institute of Technology (College/University)

Year Graduated 1990

Delete this School

School William McKinley (High School)

Year Graduated 1986

Delete this School

Professional **Add Another Job**

Employer Dunder Mifflin

Job Title Assistant to the Regional Manager

Delete this Job

Cancel **Save**

May be left blank

Adds a new interest and refreshes the page

Adds another box on the user interface with School and Year Graduated.

Lists available schools and their type

Adds another box to the user interface with Employer and Job Title.

Figure 3 - Edit Profile

Viewing a GTOOnline Profile

After logging in, a user who has already set up a profile should immediately see the View Profile screen for his or her own profile. Figure 4 shows the View Profile screen. All information from the user's profile should appear on this screen. However, the **Year Graduated** field should only be shown if the user has actually graduated and provided the graduation date for that school. The schools should be sorted by graduation date in descending order (most recent schools appear at the top). If the user has not graduated from a school then that school should appear at the top.

View GTOOnline Profile for Michael Bluth

Michael Bluth

[View Status Updates](#)
[View Friends](#)
[Log Out](#)

[Search for Friends](#)
[View Pending Requests](#)
[Edit Profile](#)

Sex Male
Birthdate 1968-06-20
Current City Scranton, PA
Hometown Beverly Hills, CA
Interests Tennis
 Watching Inception Over and Over Again
 Seinfeld

Education

School	Georgia Institute of Technology (College/University)
Year Graduated	1990

School Name	William McKinley (High School)
Year Graduated	1986

Professional

Employer	Dunder Mifflin
Job Title	Assistant to the Regional Manager

Callout 1: Search for Friends, View Pending Requests, and Edit Profile are only visible if this is the profile of the currently logged in user

Callout 2: Only show Year Graduated if field is not null

Figure 4 - View Profile

There are a number of links on the View Profile screen:

- Clicking **View Status Updates** shows the latest Status Updates from the user (see Figure 9).
- Clicking **View Friends** shows the list of friends for this user (see Figure 8).
- Clicking **Log Out** signs the user out of the system and shows the login screen.

In addition to these links, two more links appear if the profile is that of the currently logged in user (i.e., if the user is looking at someone else's profile, he or she will not see these options).

- Clicking **View Pending Requests** shows the list of friend requests that have not yet been accepted or rejected (see Figure 7).
- Clicking **Edit Profile** shows the Edit Profile screen allowing the user to make changes to his or her own profile (see Figure 3).

Requesting and Adding Friends

Social networks are all about making connections. GTOonline allows users to search for friends and connect to them. There are several steps involved in making a connection with a new friend on GTOonline:

1. A user searches for a friend based on several profile criteria including **Name**, **Email Address**, and **Hometown** (see Figure 5).
2. Then, the user submits a friend request to another user (see Figure 6).
3. Finally, the other user receives the friend request and accepts it or rejects it (see Figure 7).

A significant difference between GTOonline and Facebook is that friendship is not always reciprocal in GTOonline. (Perhaps this is a better representation of real life!) This means, for example, that just because Emily is friends with Sarah, this does not imply that Sarah is friends with Emily. In other words, in order to establish a two-way connection between Emily and Sarah on GTOonline, both users must send friend requests to each other and both requests must be accepted. If, however, only Emily sends a request to Sarah and Sarah accepts it, then Sarah will appear in Emily's list of friends, but Emily will *not* appear in Sarah's list of friends (because Sarah has not yet made a request to Emily). Technically speaking, this means that your data model must differentiate the user who requested the friendship from the one who accepted it.

The screenshot shows a web interface titled "Search for Friends". It has a green header bar with a close button (X). Below the header, there's a "Search for Friends" section with a search icon. It contains three input fields: "Name", "Email", and "Hometown". The "Hometown" field is pre-filled with "Scranton". To the right of these fields are "Close" and "Search" buttons. A callout box points to these three fields, stating "These three fields are ORed together". Below the search section is a "Search Results" section. It contains a table with two columns: "Name" and "Hometown". The "Name" column lists "Bob Vance" and "Phyllis Lapin-Vance" as blue links. The "Hometown" column lists "Scranton, PA" for both. A callout box points to the "Name" column, stating "Existing friends should not be shown in the list". Another callout box points to the "Phyllis Lapin-Vance" link, stating "Clicking on a name shows the Add Friend dialog box". A third callout box points to the "Search Results" section, stating "After clicking Search, any relevant results appear here (or on a new page)".

Figure 5 - Search for Friends

Figure 5 shows the Search for Friends screen. This screen has two parts. One part contains several fields that the user can fill in to search for friends. The user does not have to fill in all three fields. The search should return users that meet any of the criteria. That means, for example, that searching for the **Name** "John" and the **Hometown** "Scranton" will return users who have the substring "John" in any part of their **Name** OR who have the substring "Scranton" in their **Hometown** OR both.

The second part of the screen shows search results. The results should be sorted alphabetically by **First Name** and then **Last Name** and should also list the **Hometown** for the users returned. For simplicity of implementation, you may choose to divide this screen up into two separate screens: the first contains the search criteria and the second lists the results.

When the user clicks on a **Name** in the search results, the Request New Friend screen is shown (see Figure 6). This screen lists the **Name** and **Hometown** of the user who will receive the friend request.

Also, the user must provide a word or two describing the **Relationship** with the friend (e.g., co-worker, brother, sister, classmate, etc.).

Request New Friend for Michael Bluth

Request New Friend for Michael Bluth ⓘ

Name Phyllis Lapin-Vance

Hometown Scranton, PA

Relationship Co-worker

Cancel Send Friend Request

The user provides any value for the relationship

Friend is not immediately added. Instead, a request is sent to the other user.

Figure 6 - Request New Friend

From a user's profile there is a link to see any pending friend requests (see Figure 4). The Pending Friend Request screen shown in Figure 7 lists two kinds of requests: requests from other users that the current user can accept or reject, and requests that the current user has sent to other users that have not yet been accepted or rejected. Once a request has been accepted or rejected, it is removed from the Pending Friend Request screen of both the user who initiated the request and the user who accepted/rejected request. A user may also cancel a request that he or she sent before the other user has accepted/rejected it. Once a request has been cancelled, it should no longer appear on either user's Pending Friend Request screen.

Pending Friend Requests for Michael Bluth

The following people have requested to be friends with you ⓘ

Name	Hometown	Relationship	Accept	Reject
Sue Sylvester	Lima, OH	Former Coach	Accept	Reject
Will Schuester	Lima, OH	Former Director	Accept	Reject

You have requested to be friends with these people ⓘ

Name	Hometown	Relationship	Cancel
Phyllis Lapin-Vance	Scranton, PA	Co-worker	Cancel

Clicking Accept adds the user as a friend and records the current date

Clicking Reject cancels the request

Cancel a request before it is accepted or rejected

Clicking on a user's name shows his or her profile

Close

It is possible to close the window even with pending requests. Those requests should still appear the next time the user views this page.

Figure 7 - Pending Friend Requests

From a user's profile, there is a link to see the current list of friends connected to the user. This list only shows direct, first-level friends (i.e., no friends-of-friends, etc.). By default, the friends list is sorted by the **First Name**. However, the user may click on the table header to change the sorting to the **Relationship** or **Connected Since** fields. Note that the **Connected Since** field is set when the friend request is accepted, *not* when the request is originally sent.

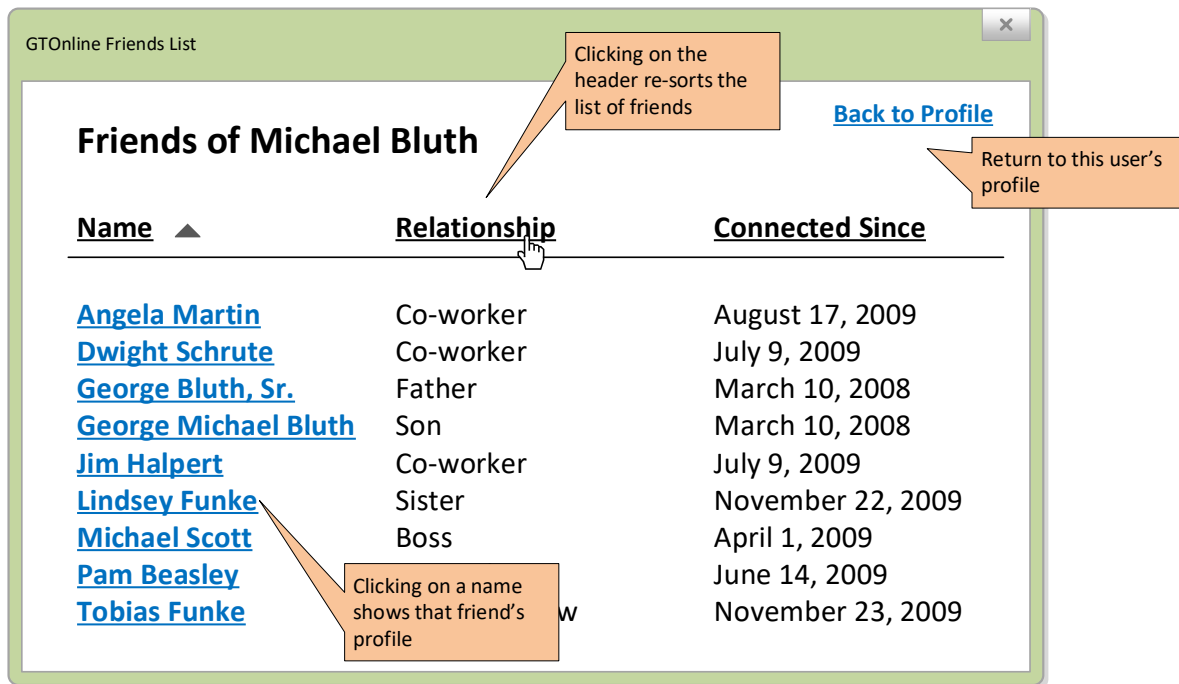


Figure 8 - View List of Friends

GTOOnline Status Updates

A social networking application would be nothing without Status Updates. So, even a lightweight social network like GTOOnline supports Status Updates. A **Status Update** is a brief snippet of text that lets a user's friends know what the user is currently doing or thinking. From the View Profile page, there is a link to view that user's most recent Status Updates (see Figure 4). The View Status Update screen is shown in Figure 9 and serves multiple purposes:

- It allows friends of a user to view the latest Status Updates from that user as well as any Comments that have been made (by any user).
- It allows a user to add new Status Updates.
- It allows users to add Comments in response to their friends' Status Updates.

Clearly, users should only be able to add Status Updates to their own profiles. Therefore, the option to add a Status Update is only shown if the Status Update screen is for the currently logged in user. Due to limited screen space, GTOOnline only shows the last three Status Updates and any comments that have been made on those Status Updates. Status Updates should be sorted in descending order with the latest update appearing at the top. The date and time of the Status Update should be shown along with the actual text of the Update.

Comments make Status Updates fun because they allow users to make (smart) remarks about the Status Updates of their friends. Comments are always associated with a particular Status Update. As seen in Figure 9, an Add Comment button appears next to each Status Update allowing the currently logged in user to comment on their friends' Status Updates. Comments should be sorted in ascending order with the most recent comment (for each Status Update) appearing at the bottom. There is no limit to the number of Comments allowed on a particular Status Update. It is also possible for a user to comment on his or her own Status Update.

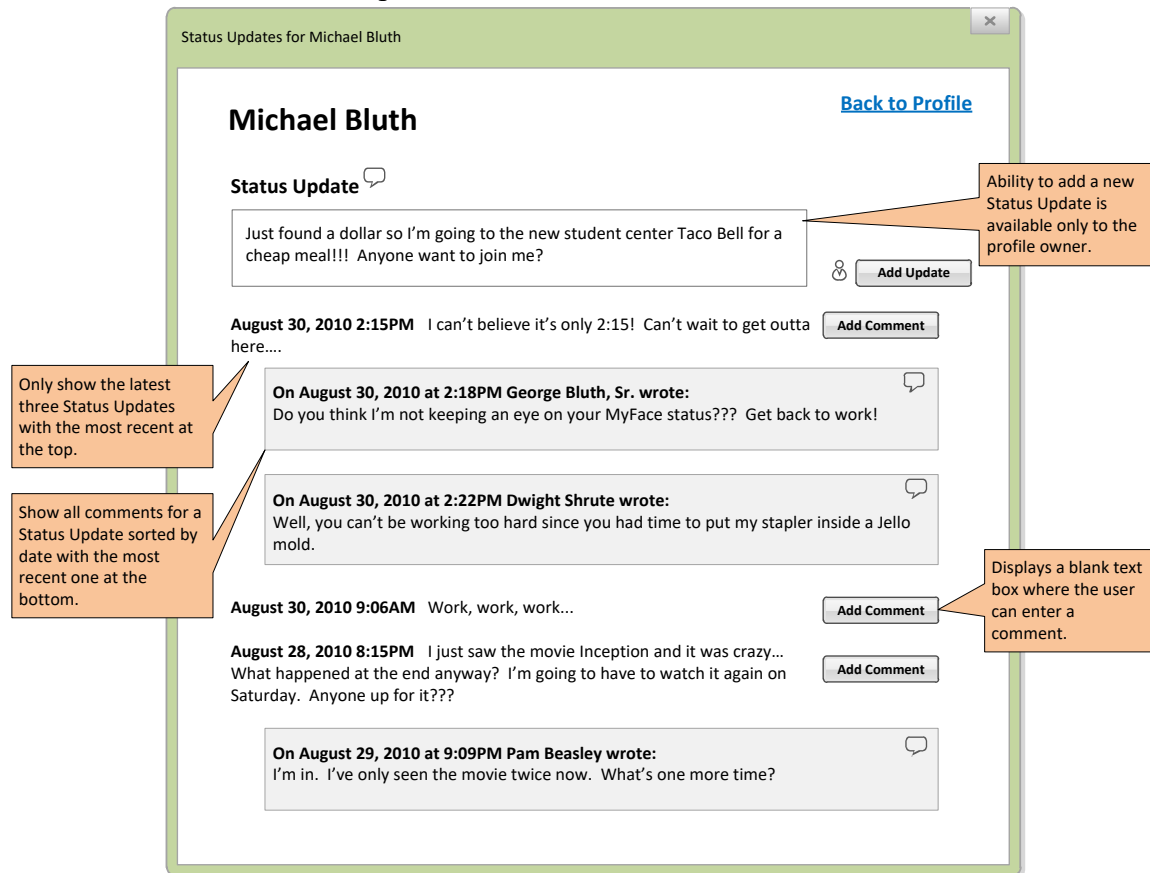


Figure 9 - View Status Updates

Reports

An admin user will have a “View Reports” link on the “View Profile” form.

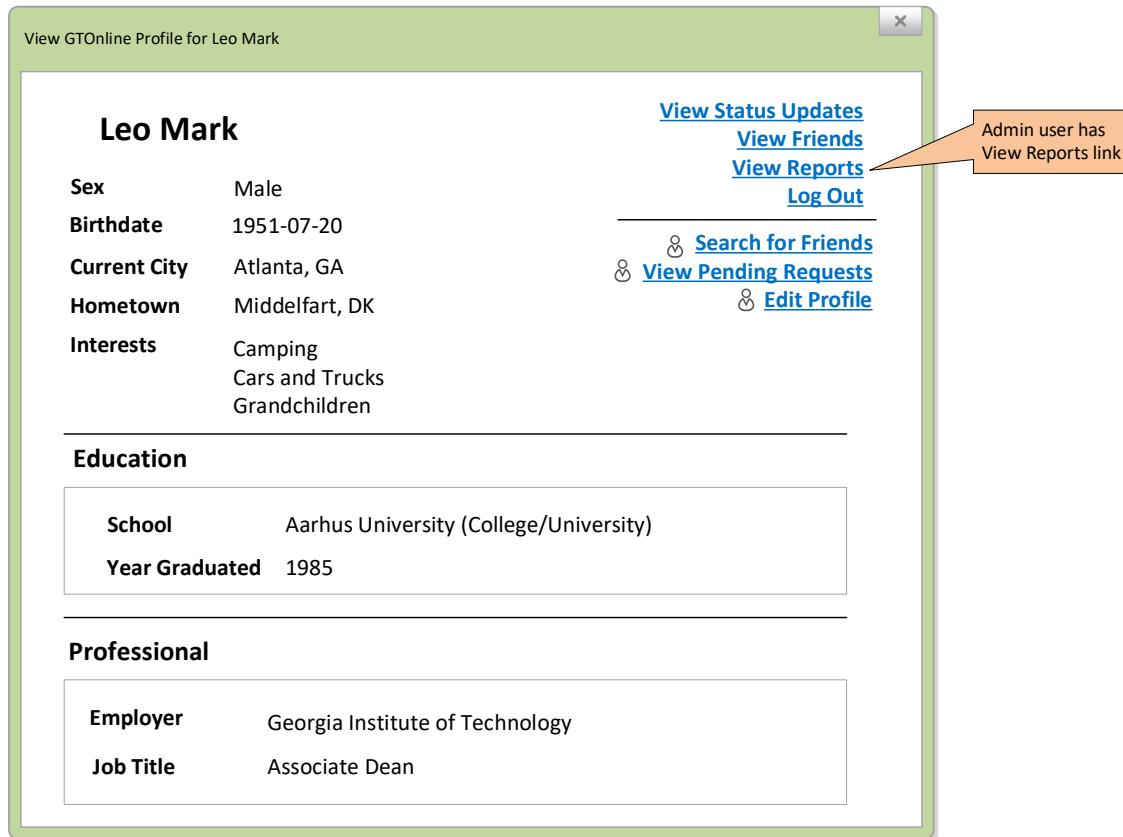


Figure 10 - View Profile for Admin user has View Reports Link

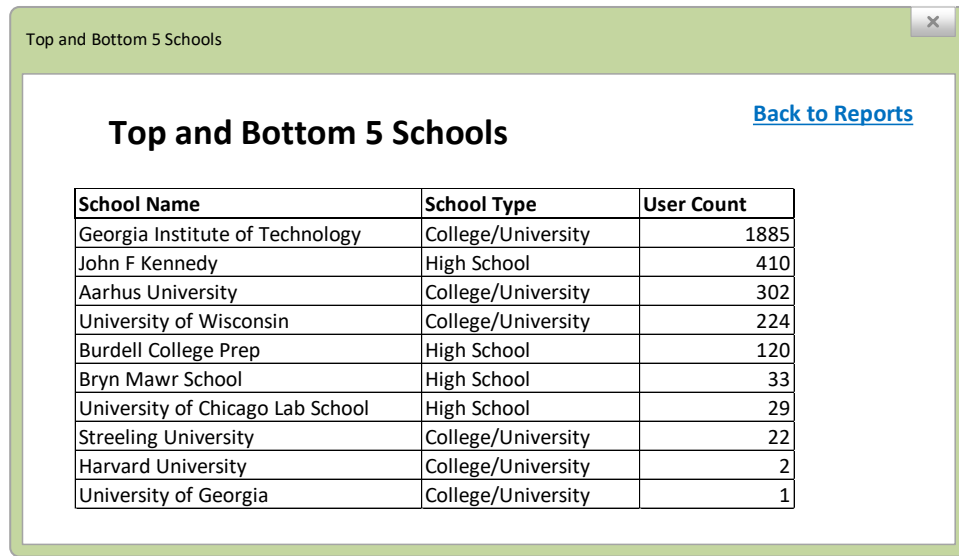
Clicking “View Reports” will show a listing of reports. Clicking on the link for a report will open it.



Figure 11 - List of reports

Top and Bottom 5 Schools

This report will show the top five schools, and bottom five schools, based on the distinct count of users who have listed that school. The results should show the school name, school type, count of users, and ordered by the count of users descending.

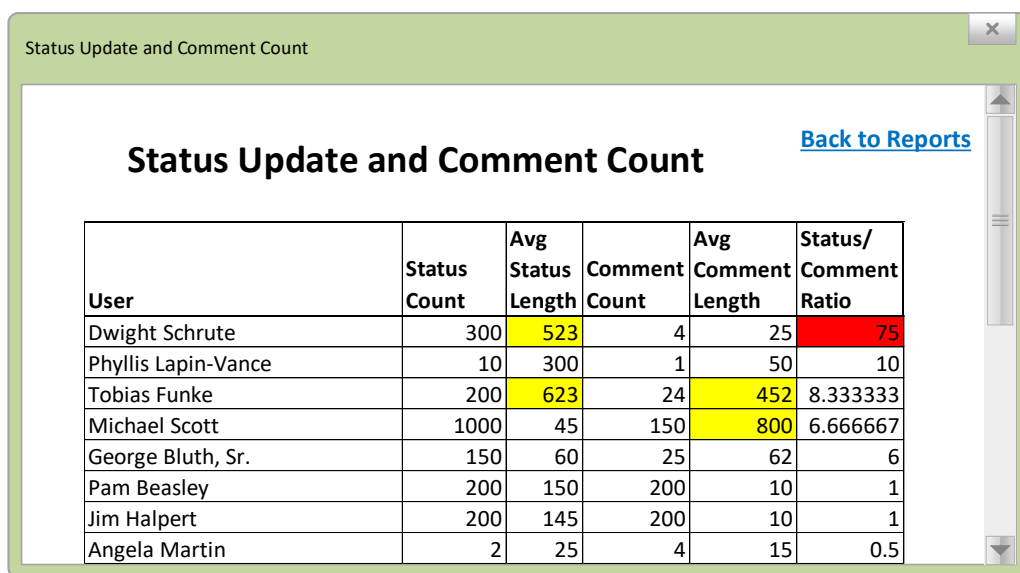


School Name	School Type	User Count
Georgia Institute of Technology	College/University	1885
John F Kennedy	High School	410
Aarhus University	College/University	302
University of Wisconsin	College/University	224
Burdell College Prep	High School	120
Bryn Mawr School	High School	33
University of Chicago Lab School	High School	29
Streeling University	College/University	22
Harvard University	College/University	2
University of Georgia	College/University	1

Figure 12 - Top and Bottom 5 Schools

Status Update and Comment Count

For each user, calculate the count of status updates, the average length of status updates, the count of comments, the average comment length, and the ratio of status updates to comments (count status updates / count comments), ordered by ratio descending. If a user's average length for status updates or comments exceeds 400, highlight the cell with that value with a yellow background. If a user's status to comment ratio is greater than or equal to 50, highlight the cell with that value with a red background.



User	Status Count	Avg Status Length	Comment Count	Avg Comment Length	Status/Comment Ratio
Dwight Schrute	300	523	4	25	75
Phyllis Lapin-Vance	10	300	1	50	10
Tobias Funke	200	623	24	452	8.333333
Michael Scott	1000	45	150	800	6.666667
George Bluth, Sr.	150	60	25	62	6
Pam Beasley	200	150	200	10	1
Jim Halpert	200	145	200	10	1
Angela Martin	2	25	4	15	0.5

Figure 13 - Status Update and Comment Count

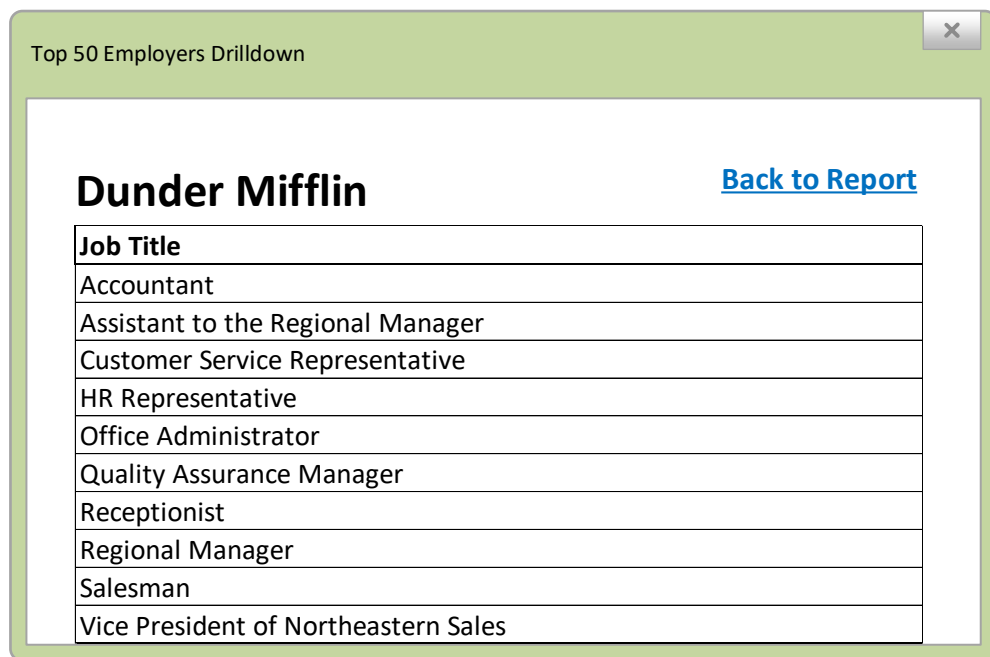
Top 50 Employers

Calculate the number of job titles associated with each employer, listing the employer name, and the count, ordered by count descending. The job count should link to a drill-down report listing all distinct job titles associated with that employer, ordered by title ascending.



Employer	Job Count
Dunder Mifflin	127
Bluth Company	90
Omni Consumer Products	88
InGen	75
Wayne Enterprises	62
Silicon Graphics	60
Kwik-E-Mart	57
WebTV	52
USRobotics	47
Megadodo Publications	42

Figure 14 - Top 50 Employers

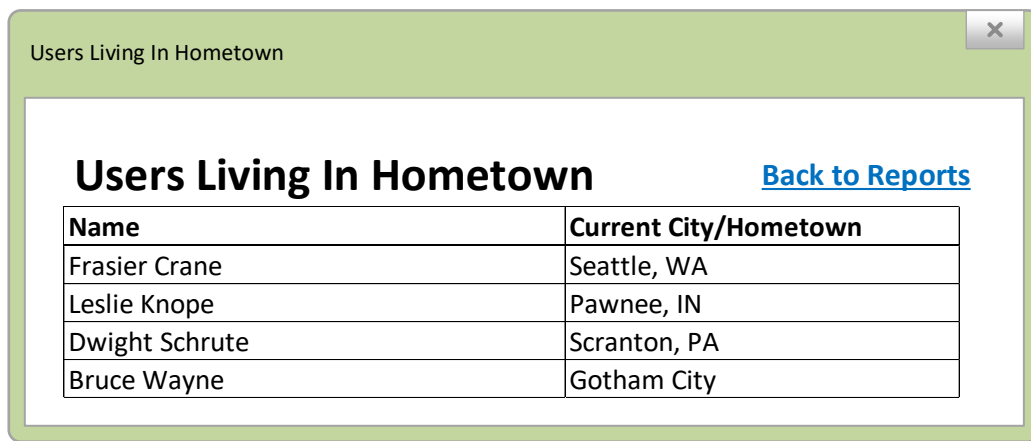


Job Title
Accountant
Assistant to the Regional Manager
Customer Service Representative
HR Representative
Office Administrator
Quality Assurance Manager
Receptionist
Regional Manager
Salesman
Vice President of Northeastern Sales

Figure 15 - Top 50 Employers Drilldown

Users Living In Hometown

This report will list users whose current city and hometown are identical. The user's name and current city/hometown will be displayed, order by user last name ascending.

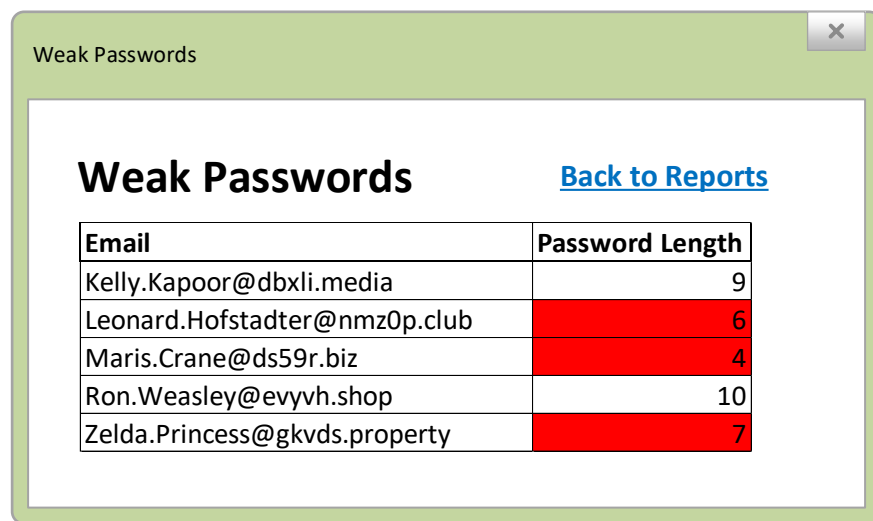


Name	Current City/Hometown
Frasier Crane	Seattle, WA
Leslie Knope	Pawnee, IN
Dwight Schrute	Scranton, PA
Bruce Wayne	Gotham City

Figure 16 - Users Living In Hometown

Weak Passwords

Users whose passwords are less than or equal to 10 characters will be shown on this report. The user's email and password length should be displayed, with rows sorted by email address ascending. If a user's password is less than or equal to 7 characters, highlight the cell with that value with a red background to indicate users who should change to a longer password.



Email	Password Length
Kelly.Kapoor@dbxli.media	9
Leonard.Hofstadter@nmz0p.club	6
Maris.Crane@ds59r.biz	4
Ron.Weasley@evyvh.shop	10
Zelda.Princess@gkvds.property	7

Figure 17 - Weak Passwords